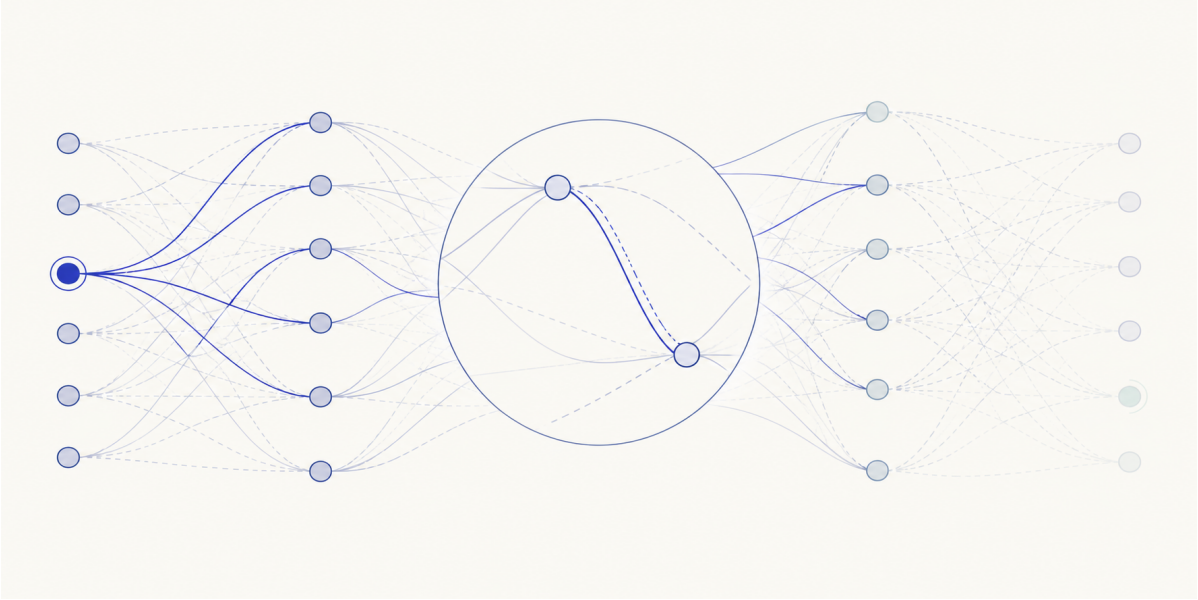
What an ablation can resolve

Measuring the limits of intervention-based interpretability in small graph neural networks, where exact ground truth is affordable.

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Summary

I measured how changing each hidden channel in a graph neural network affected every channel in the next layer, then compared those exact effects with a gradient-based approximation. Across 30 trained models, the median rank correlation was **0.9814**, sign agreement was **95.64%**, and the approximation recovered **87.50%** of the largest effects in a fixed-size shortlist. Small models made it possible to check the approximation against the interventions it was meant to approximate.

That agreement was the strongest result. The controls also changed what I thought the interventions could tell me. Learned feature directions moved the model's output **2.64 times as much** as random directions at the same intervention size. But they had a much smaller advantage in predicting *which classes* an intervention would affect. Much of that second signal survived replacing the learned directions while preserving how the intervention varied across graph nodes.

Some patterns of influence between layers also recurred across training runs. The weaker effects were consistently measurable within a model but less conserved between models. And channels matched by their cross-layer connections did not meet the criterion for matching in their effects on output classes. Agreement between two measurements, similarity between trained models, and similarity of function turned out to be three different results.

Question	Main result	Scope
Does first-order attribution reproduce exact channel-pair effects?	Median Spearman correlation **0.9814**, positive in 30/30 models	Exact interventions in small models
Does the original channel basis outperform random rotations?	Paired contrast **−0.0067**, inside the registered ±0.05 band	Pooled result, substantial variation across model groups
Do learned directions better organize effects by class?	Advantage **+0.0359**, below the 0.05 threshold	Random directions matched for intervention size at each node
Do learned directions move outputs more?	**2.64×**, with all 30 models above a 1.25× threshold	Top-impact features, compared at matched displacement
Does cross-layer connection structure recur across runs?	Correlation gain **+0.0985**, positive in 30/30 models	Stronger effects contribute most to this result
Do the matched channels have more similar classwise effects?	Cosine-similarity gain **+0.0087**, below the 0.05 threshold	This output-based description of function

Here, each checkpoint is a separately trained model. The counts summarize the direction of a comparison within models, not independent datasets or individual significance tests.

Epistemic status. These are exploratory studies on the same 30 previously inspected development checkpoints, in four dataset–architecture groups with widths of 32 or 64. Protocols and thresholds were frozen before each full evaluation, with the previews and repaired rerun described below. They are not new holdout replications. I am confident about the reported comparisons in this setting. Their generalization to larger models, other intervention definitions, or language models remains open. The companion paper, *What an Ablation Resolves*, is **in preparation**.

This continues the work in Can a component's intervention footprint reveal its function?, which grew out of my graduate research and my return to interpretability as an independent researcher. The decision records and figure measurements are public.

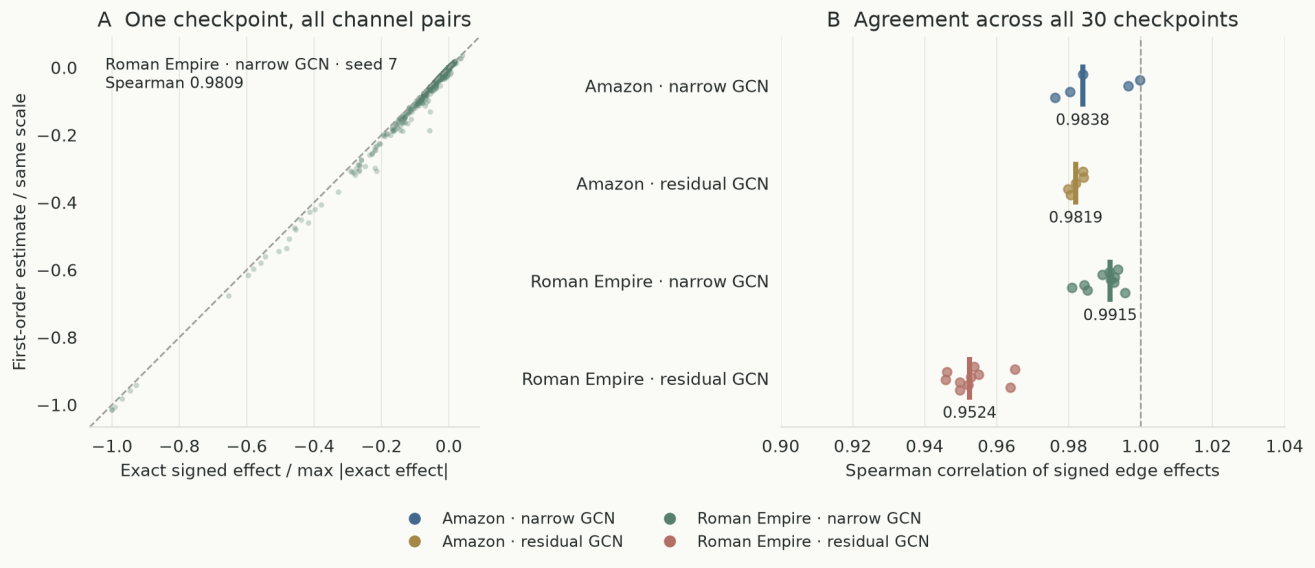


Figure 1. First-order attribution checked against exact interventions. Left: all 1,024 channel pairs in the checkpoint nearest the pooled median correlation, Roman Empire / narrow GCN, seed 7. Both axes use the same scale factor. Right: all 30 checkpoint correlations, with bars marking the four group medians. The example was selected by proximity to the median, not by best agreement. This checks the signed aggregate effect defined in Eq. 4 below.

The intervention itself can explain part of the pattern

In the first post, I described a channel by the predictions that changed when I intervened on it. Channels with overlapping intervention footprints tended to affect similar classes. That relationship survived a prospective test on new datasets, but it did not identify pairs whose learned repairs combined more favorably.

This left a more basic question: how much of the footprint's apparent organization came from the learned channel, and how much came from the intervention? Replacing a channel changes several things together. It chooses a direction in the hidden representation, determines the size of the change, and determines how that change is distributed across examples. A useful control has to separate those choices.

The models here predict a class for each node in a graph. Each node has a vector of hidden activations, and a hidden channel is one coordinate of that vector across nodes. Graph layers combine information from connected nodes, so a local change can affect predictions elsewhere. These small models let me examine every source–target channel pair and run exact interventions alongside the approximation. That is the practical reason to use them for this question.

I started by keeping the intervention rule fixed while changing what counted as a component.

Mean replacement lets me test channels against other directions

The first post replaced one channel's activation with its training-node mean. The same operation can be applied along any unit direction u in the hidden layer:

$$h'_v = h_v - \langle h_v, u\rangle u + \langle \bar{h}_{\text{train}}, u\rangle u, \qquad \|u\|_2 = 1. \tag{1}$$

Here h_v is the complete hidden vector at node v. The second term removes its component along u, and the third restores that component at its training mean. Everything orthogonal to u stays unchanged. The remaining model reads the full updated vector h'_v. Choosing a coordinate axis for u gives the channel intervention from Eq. 1 of the first post.

For example, suppose a layer has two coordinates and u points halfway between their axes. Equation 1 changes both coordinates together, while preserving the component perpendicular to that diagonal. Comparing this with a coordinate intervention tests whether the original axes are especially informative for the measurement. It does not retrain or rotate the model's weights.

A **footprint** records the resulting change in output probabilities at each node. I use total variation distance, half the sum of the absolute changes in class probabilities. For a two-class prediction moving from (0.8, 0.2) to (0.6, 0.4), the footprint value is 0.2 even though the predicted class stays the same. Summing this distance over evaluated nodes gives the footprint's **mass**.

A separate **classwise profile** records the mean KL divergence in predictions for each true class. Two interventions have similar profiles if they disturb the same classes by similar relative amounts. The alignment score asks whether interventions with overlapping footprints also have similar classwise profiles:

$$\begin{aligned}&\text{alignment improvement}\\&\quad = \rho_S(\text{footprint overlap}, \text{classwise-profile similarity})\\&\qquad - \rho_S(\text{baseline similarity}, \text{classwise-profile similarity}).\end{aligned} \tag{2}$$

The baseline uses effect magnitude and graph-node-role summaries. Equation 2 therefore asks what footprint overlap adds beyond those simpler descriptions. It is not a measure of total output movement. This distinction matters when the direction controls produce a modest alignment result but a much larger amplification result.

I use **cell** below for one dataset–architecture combination. Unless stated otherwise, a headline number is the median of the 30 checkpoint values. Cell ranges show the range of the four cell medians.

Random rotations preserve the alignment signal, with large differences between cells

The original channel axes were not consistently privileged by Eq. 2. Five random orthonormal bases produced pooled alignment improvements between **+0.2580 and +0.2846**, each positive in 30/30 checkpoints. The channel basis gave **+0.2415**, positive in 29/30.

The registered comparison paired each checkpoint's channel result with the median of its five rotations. The pooled channel-minus-rotation contrast was **−0.0067** among active directions and **−0.0281** when all directions were retained. Both fell inside the ±0.05 band used for the pooled tie verdict. A signed permutation of the channels also reproduced the underlying footprints, providing a check that changing their labels and signs did not alter the intervention. Basis-comparison record.

The pooled result conceals an important spread:

Cell	Channel minus random-rotation alignment
Amazon Computers / narrow GCN	**−0.2815**
Amazon Computers / residual GCN	**+0.0390**
Roman Empire / narrow GCN	**+0.1037**
Roman Empire / residual GCN	**−0.0145**

Two cells lie outside the band, in opposite directions. Rotations clearly outperform the channel basis in Amazon's narrow GCN. The pooled tie is neither a mathematical invariance nor a claim that every model ties, and individual footprints can change substantially under rotation.

What survives is more limited and useful: the alignment signal is not exclusive to the original neuron coordinates. Rotated directions within the same layer can carry it too. This endpoint alone does not establish that those coordinates identify a privileged set of functional components.

Principal-component directions initially looked less successful, at **+0.1727**. But their interventions also moved the hidden state much less. The ratio of channel to PCA displacement ranged from **2.09× to 50.62×** across cells. That makes the comparison ambiguous: changing the basis also changed the intervention's size. Even the random rotations were not perfectly size-matched, with pooled displacements 2.7%–7.4% below the channel basis. The next experiment controlled that variable directly. Measurements and aggregation.

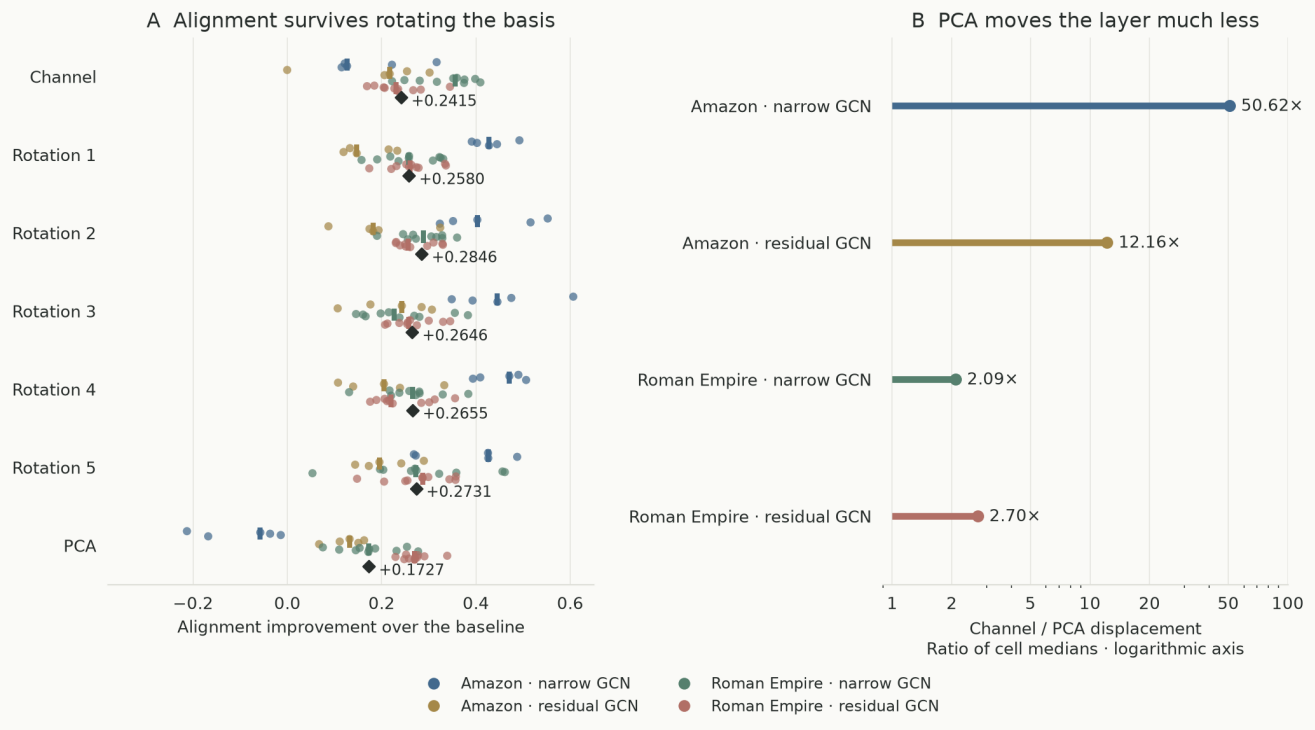


Figure 2. Left: alignment improvements for channels, five random bases, and PCA directions. Dots are checkpoints, colored bars cell medians, and black diamonds pooled medians. Right: ratios of cell-median intervention sizes show the PCA confound. A lower alignment score under a much smaller intervention does not isolate an effect of direction.

Matching displacement separates output amplification from classwise alignment

Learned directions did amplify output movement. Their advantage in classwise alignment was much smaller, and preserving the intervention's pattern across nodes mattered more for that endpoint.

I fitted sparse dictionaries to hidden activations. Each feature has a scalar activation $z_k(v)$ and a decoder direction d_k. Replacing the feature's activation with its training mean changes the original hidden state by a vector along d_k, while preserving the dictionary's reconstruction residual. Let $\delta_k(v)$ be the feature activation minus its training mean. The learned and random-direction edits are:

$$\Delta h_v^{\text{learned}} = -\delta_k(v) d_k, \qquad \Delta h_v^{\text{random}} = -\delta_k(v) \|d_k\|_2 u, \qquad \|u\|_2 = 1. \tag{3}$$

The model's next layer reads the original hidden state plus the edit in Eq. 3. The two edits have the same length at every node. Only the direction changes. A second control keeps the learned direction but shuffles the coefficients across nodes, changing where the larger and smaller interventions occur. I call that the **node-profile shuffle**. The records call it a support control, but the object being shuffled is the full displacement profile, not just an on/off mask. Mean replacement can make a small nonzero edit even where the original sparse feature is inactive.

The worst relative size mismatch between the learned edit and its matched random-direction control was $\mathbf{1.4 \times 10^{-7}}$, well inside the 1% gate. That check matters because the PCA comparison had shown how intervention size could confound the conclusion.

Changing direction reduced alignment by **+0.0359**, positive in 28/30 checkpoints, with cell medians from **+0.0112 to +0.0441**. Shuffling the node profile reduced it by **+0.1146**, positive in 30/30, with cell medians from **+0.0167 to +0.1510**. The latter cost was about three times larger in the pooled comparison, though not in every cell. When I matched the number of active directions across conditions, the corresponding values were **+0.0247** and **+0.1097**. Their cell ranges were **−0.0302 to +0.0331** and **+0.0715 to +0.2206**, respectively. The direction comparison was then positive in only two of four cells, while the node-profile comparison remained positive in all four. Neither direction subset cleared the registered +0.05 threshold. Matched-null record.

That does not say learned directions are inconsequential. Equation 2 rewards organization beyond a magnitude baseline. To ask about output movement itself, I compared the total footprint mass produced by the two equal-length edits in Eq. 3.

The registered **top-impact** subset selects ten features per dictionary by their summed absolute coefficients on evaluated nodes. These coefficients are shared by the learned and null conditions, so selection does not use the output-amplification ratio. In this subset, learned directions produced **2.64×** the output movement of their matched random directions. Across all live features the ratio was **1.36×**. Both subsets exceeded the prespecified **1.25×** threshold in all 30 checkpoints. Cell medians ranged from **2.10× to 4.24×** for the top-impact subset and **1.33× to 1.86×** for all live features. These summarize paired ratios in log space, rather than dividing unrelated pooled means. Amplification record.

One checkpoint had been previewed before freezing that study. It was the weakest top-impact result, **1.74×**. Excluding it raises the pooled ratio to **2.66×** across the remaining 29, so that preview does not drive the headline.

The two endpoints answer different questions. The learned direction helps determine how much the output moves. The intervention's distribution across nodes explains more of the measured alignment with classwise effects. Neither endpoint by itself tells me whether the change is useful for a task.

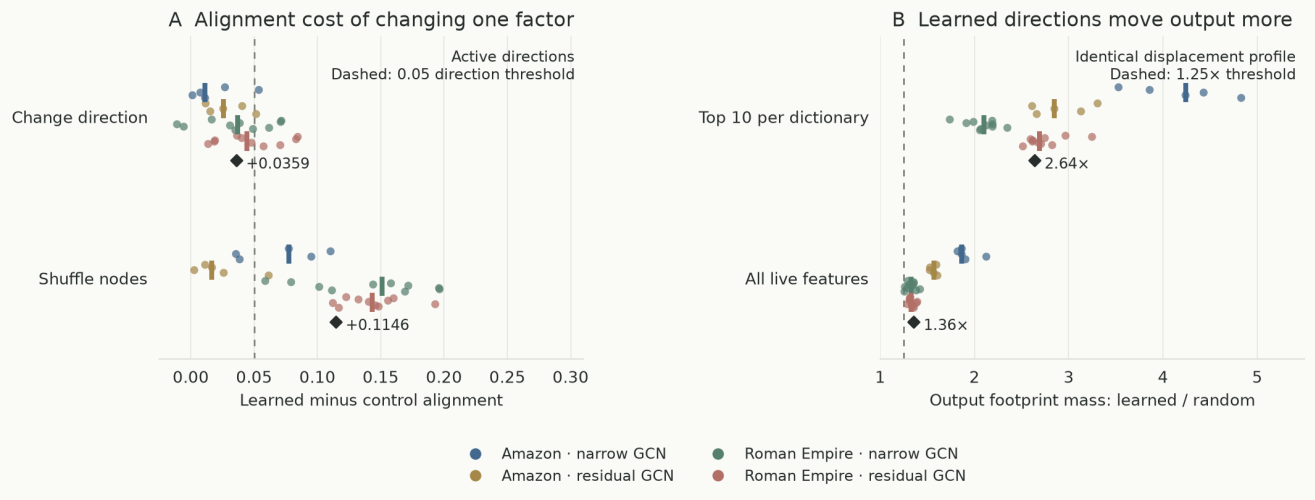


Figure 3. Left: losses in alignment when changing direction or shuffling the node profile, using the active-direction subset. Right: amplification of output movement at matched displacement. The dashed lines are the respective registered thresholds, +0.05 and 1.25×. Both panels show checkpoint values and cell medians, but their units and questions differ.

Two apparent advantages disappeared under the matched control

The matched null also overturned two earlier interpretations of the sparse features. Those interpretations were tempting: their output effects looked more class-specific and much more localized across graph nodes than effects from random rotations.

"Class-specific" here meant lower entropy of the distribution of effects across output classes. It was a proposed indicator of monosemanticity, not a direct demonstration that a feature represented one concept. "Localized" meant a lower participation ratio across nodes, with smaller values indicating concentration on fewer nodes.

The original class-entropy improvement over rotations was **+0.1278**, positive in all 30 checkpoints. Against the displacement-matched random directions, it became **+0.0104**, positive in 20/30, with cell medians ranging from **−0.0008 to +0.0508**. The claim of roughly ninefold greater localization also disappeared. The matched participation-ratio gain was **−0.0006**, positive in only 11/30, with cell medians from **−0.0036 to +0.0030**. Original record with retraction addendum.

The absolute levels make the problem clearer. Learned and matched-random effects had class entropies of **0.7321 versus 0.7465**, and participation ratios of **0.0417 versus 0.0408**. A random direction, given the learned feature's displacement profile, reproduced almost all of the apparent localization and most of the class-specificity advantage. The pooled levels and paired gains use different aggregation operations, so subtracting those two entropy medians does not recover the +0.0104 paired gain.

These were two of the strongest-looking results in the initial comparison. I withdrew their interpretation as evidence for the learned directions. The intervention profile could create the effect without those directions, and the first control had failed to hold that profile fixed. That is a substantive correction, not just a smaller effect size.

First-order attribution closely reproduces exact cross-layer effects

The next measurement retained information that the unsigned output footprint discarded: which downstream channel changed, and in which direction. It also allowed a direct check of attribution fidelity.

Take source channel i in layer s and target channel j in the next layer t. I apply the mean-replacement intervention to source channel i across nodes, run the remaining computation, and compare target activations with the unmodified run:

$$\begin{aligned} e_{ij}(v) &= h_{t,j}^{(-i)}(v) - h_{t,j}(v), \\ E_{ij} = \sum_{v \in V_{\text{eval}}} e_{ij}(v), &\qquad S_{ij} = \sum_{v \in V_{\text{eval}}} |e_{ij}(v)|. \end{aligned} \tag{4}$$

The field $e_{ij}(v)$ records the effect at each evaluated node. Its signed sum E_{ij} can cancel positive and negative effects. Its strength S_{ij} counts their absolute sizes. A “cross-layer edge” below is this measured relation between channels, not an edge between graph nodes and not necessarily a direct weight in the network.

For example, an intervention might increase a target channel at one node and decrease it at another. Its signed aggregate could be near zero even though the strength is substantial. I use the signed aggregate for the main attribution comparison and the strength for questions about concentration.

First-order attribution estimates Eq. 4 from the model’s derivatives at the unmodified activation. If μ_i is the source channel’s training mean, its predicted signed effect is:

$$\widehat{E}_{ij} = \sum_{a \in V} \big(\mu_i - h_{s,i}(a)\big) \frac{\partial \left[\sum_{v \in V_{\text{eval}}} h_{t,j}(v)\right]}{\partial h_{s,i}(a)}. \tag{5}$$

The derivative includes the graph layer’s mixing of information between nodes. The sum over a covers the nodes where the source is edited, while the target sum covers the evaluation nodes. This estimates a finite mean-replacement intervention using a local linear approximation. Exact evaluation runs one intervention per source channel. The approximation uses one backward pass per target channel. I did not benchmark a universal speedup between them.

For these exact interventions, agreement was high:

Fidelity measure	Median across checkpoints	Range of cell medians
Spearman correlation of signed effects	**0.9814**	**0.9524–0.9915**
Spearman correlation of absolute signed effects	**0.9708**	**0.8614–0.9972**
Sign agreement	**95.64%**	**90.48%–99.51%**
Recovery of the largest-effect shortlist	**87.50%**	**73.44%–95.31%**

The shortlist contains the largest 32 or 64 entries of the whole channel-pair matrix, matching the model's hidden width. Sign agreement includes entries where both effects are zero. All 30 checkpoints were evaluable. The magnitude range above corrects an archival table that understated its maximum. Exact-effects record and clarification.

The highest cell-median signed correlation was the Roman Empire narrow GCN's **0.9915**, and the lowest was its residual GCN's **0.9524**. The former uses a ReLU path, while the latter includes GELU and LayerNorm. That ordering is consistent with greater departures from the linear approximation, although these architectures differ in other ways too.

This is the sense in which Figure 1 provides ground truth: I can check the approximation against the precise intervention it claims to estimate. It does not establish that channel pairs are the uniquely correct units of computation, or that their aggregate effects explain the model's behavior.

A few channel pairs carry disproportionate mass, but their effects span many nodes

Cross-layer influence was concentrated across channel pairs and broad across graph nodes. Those are different axes of the same measurement.

The strongest **5% of channel-pair edges carried 24.68%** of total strength, with cell medians from **18.93% to 35.09%**. The median concentration factor relative to a uniform allocation was **4.86×**. This is disproportionate concentration, not evidence that only 5% of edges matter: the remaining pairs still carry most of the total mass.

Across graph nodes, the high-strength edges were much more spread out. Their normalized participation ratio was **0.5434**, with cell medians **0.5184–0.6133**, and normalized entropy was **0.9609**, with cell medians **0.9549–0.9692**. Both measures increase as influence spreads across nodes. These values support broad influence, not literal uniformity at every node. Cross-layer structure record.

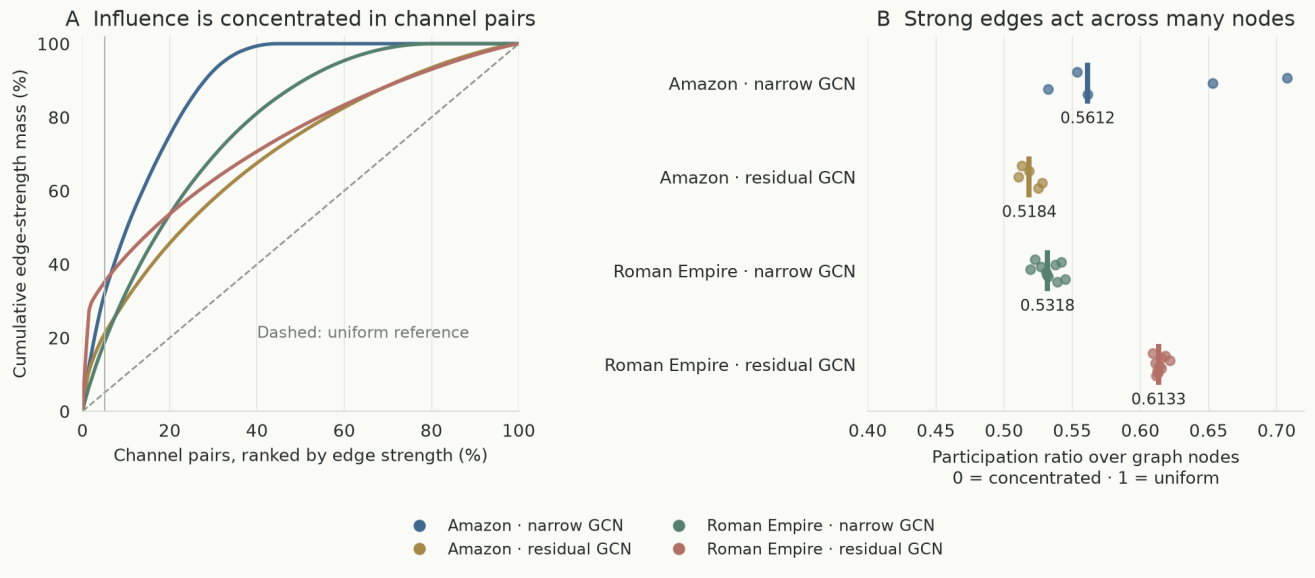


Figure 4. Left: cumulative edge-strength curves, taking cell medians after summarizing each checkpoint. The diagonal is a uniform reference, not a statistical null. Right: node-level participation ratios for high-strength edges. Concentration across channel pairs can coexist with broad effects across nodes.

There is deliberately no permutation test for the sorted-mass curve. Permuting the matrix preserves the multiset of strengths, so it preserves the curve exactly. Such a control cannot distinguish structured concentration from anything else. A uniform reference describes concentration without pretending to test its origin.

I also checked what is lost by adding separate interventions. For each target channel, I compared the sum of its changes under individual source-channel ablations with its actual change when all source channels were ablated together. The absolute discrepancy, summed over nodes and divided by the joint intervention's absolute effect, had a median of **22.67%**. Cell medians ranged from **6.59% to 32.60%**.

That discrepancy is substantial. Even very accurate single-edge attribution does not make simultaneous interventions additive. The 22.67% is the measured relative error of this particular summation, not a universal ceiling on every possible circuit account. Any account that uses these edges additively has to confront it.

Stronger connection patterns recur across training runs

After allowing channel identities to differ across runs, cross-layer edge matrices were more similar than matched controls. The active-channel correlation gain was **+0.0985**, positive in all 30 checkpoints and all four cells, above the registered +0.05 threshold. But obtaining a meaningful comparison required fixing the control first.

Channels have no stable index across independently trained models. Channel 7 in one run need not correspond to channel 7 in another. I therefore matched source and target channels to make their edge matrices comparable. The original proposed null compared the optimized match with a random assignment.

That comparison gave the real matrices a selection advantage that the control did not receive. In a check using two independent noise matrices, matching produced a correlation of about **0.37**, while a random assignment gave about **0.00**. The procedure could report apparent correspondence where none had been planted.

The repaired null gave the surrogate matrices the **same matching procedure**. Row-shuffled surrogates preserved edge values and row masses while disrupting their arrangement. Under the corrected comparison, noise controls gave gains around **−0.004 to +0.006**, while planted correspondence gave **+0.51 to +0.55**. The full study then measured the gain over these identically matched surrogates. Seed-stability record.

The real result varied considerably by cell:

Cell	Active-channel correlation gain
Amazon Computers / narrow GCN	**+0.0936**
Amazon Computers / residual GCN	**+0.1083**
Roman Empire / narrow GCN	**+0.0320**
Roman Empire / residual GCN	**+0.2471**

The weakest cell falls below +0.05 on its own. The registered rule required the pooled median to clear the bar and at least three cell medians to be positive, not every cell to exceed the bar.

Including dead channels raised the pooled gain to **+0.1351**. The all-channel to active-channel gain ratios were **0.96× and 1.00×** in the two residual cells without dead channels, compared with **1.74× and 2.13×** in the two narrow-model cells that had them. Shared inactivity can make models look more alike, which is why I emphasize the active-channel result.

The main comparison used Pearson correlation, which is more sensitive to larger entries. A rank-based variant gave only **+0.0297**, below the threshold. This suggested that conservation was stronger among large effects. It motivated testing strong and weak edges separately.

Weak effects agree within a model but are less conserved across runs

The stratified comparison supported that distinction. Among active channels, cross-seed conservation gains were **+0.0735** in the stronger half of nonzero edges and **+0.0320** in the weaker half. The head was positive in 26/30 checkpoints and the tail in 22/30. Head cell medians ranged from **−0.0089 to +0.2413**, so strong-edge conservation was not universal either. Tail cell medians ranged from **−0.0039 to +0.0477**, all below the +0.05 bar.

Weak edges nevertheless showed strong within-model agreement between exact ablation and first-order attribution. This endpoint subtracts agreement with shuffled attribution from actual exact-versus-approximate agreement. Its pooled gains were **+0.9807** for the head and **+0.9792** for the tail, each positive in 30/30 checkpoints and all four cells. The corresponding cell ranges were **+0.9580 to +0.9909** and **+0.9388 to +1.0192**. A difference of correlations can exceed one. These are **agreement gains**, not raw correlations like the 0.9814 in Figure 1. Tail-study record and correction.

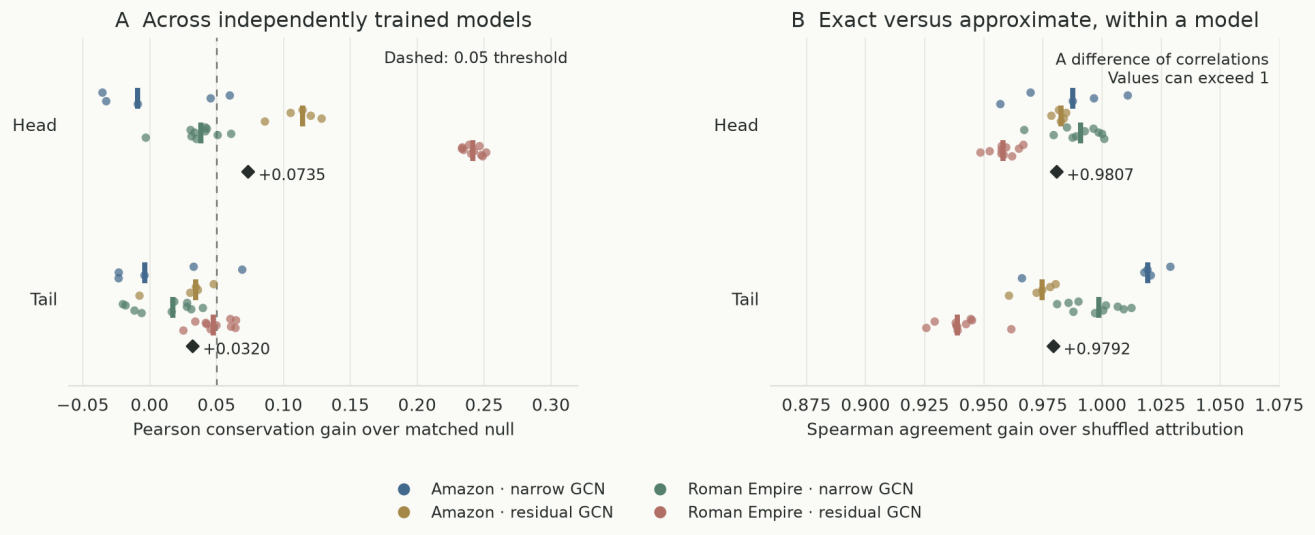


Figure 5. Head and tail are the stronger and weaker halves of nonzero edges. Left: cross-seed conservation gain. Right: within-model exact/first-order agreement gain over shuffled attribution. The right-hand numbers are differences of Spearman correlations. Consistent measurement within one model does not imply conservation between trained models.

The tail therefore contains structure that both procedures can detect within a checkpoint, while failing the program's criterion for conservation across checkpoints. That supports the recorded verdict of a **run-specific tail**. It does not establish what that structure represents or whether it matters for behavior. Both measurements use the same trained model, and their agreement is not proof that all possible sources of noise have been eliminated.

This result also has weaker provenance than a first-look test. The first version failed its coverage gate: only **24/30** checkpoints were evaluable, against a requirement of 28. In Amazon's narrow GCN, 58% of matrix entries were exactly zero, which broke the planned quartile stratification. I saw the descriptive results before freezing the repair.

The second version used halves of nonzero edges, retaining the existing minimum of 64 entries per stratum, thresholds, and decision rule. It is a **repaired-instrument rerun**, not a fresh preregistration. The all-channel analysis gave the same qualitative verdict. The result bounds conservation at this coarse split, not at a precisely identified cutoff within the weak-edge tail.

Matching connection patterns did not establish matching function

The channels paired by edge geometry had only a small advantage in classwise output similarity. The active-channel cosine-similarity gain over a mass-stratified assignment control was **+0.0087**, positive in 22/30 checkpoints. The all-channel gain was **+0.0130**, positive in 27/30. Both fell below the +0.05 threshold.

Active-channel cell medians ranged from **+0.0008 to +0.0152**, and all-channel medians from **+0.0035 to +0.0411**. Every cell median was positive, but neither subset met the rule for meaningful functional correspondence under this descriptor. Functional-comparison record.

The absolute similarity was already high: **0.7474** for matched channels versus **0.7383** for the control. These levels pool all pairs and both axes, while +0.0087 is the median of the paired gains computed within each checkpoint. Subtracting the pooled levels gives 0.0091, close to +0.0087 by coincidence rather than because it reproduces the calculation. These classwise KL profiles are nonnegative, and many channels produce broadly similar profiles. There was only about 0.26 of headroom to perfect cosine similarity. A descriptor that removes shared variation, such as a whitened profile, might distinguish functions better, but that would be a new test.

I therefore would not use the edge match to claim that channels are functionally interchangeable. It establishes recurring connection geometry under the matching procedure. It does not establish equivalent computations, transferable circuits, or equivalent repairs. Those require their own measurements, just as the repair result in the first post did.

Two control rules changed the conclusions

To isolate direction, match the displacement at every intervened position. A random direction is not enough if it changes the amount or distribution of the intervention. Equation 3 held both fixed. That exposed the PCA size confound, removed the two apparent sparse-feature advantages, and still left a clear output-amplification result. The control made the surviving positive claim more specific.

Give the control the same selection procedure, and check that the statistic can distinguish it. Optimized channel matching against an unoptimized assignment rewarded optimization even on noise. Shuffling a matrix before computing its sorted-mass curve changed nothing that the statistic could observe. These are different failures, but both can be caught before interpreting a model result.

I keep those checks alongside the scientific outcomes. The public seven-entry instrument-log extract covers the direction studies, including unequal active-channel counts and a checkpoint-provenance failure caught before amplification was computed. The cross-layer matching and concentration checks are documented in their separate decision records. The extract is not the complete log for the program.

Accurate attribution is a starting point for the next experiment

These experiments give me a measured basis for using first-order attribution in this setting: I know how closely it reproduces the exact channel-pair intervention, how concentrated those effects are, and how far their sum departs from a simultaneous intervention. I also know which similarities survive across training runs and which functional claim those similarities did not support.

The next question I want to answer is whether **amplification at matched displacement predicts a useful task effect**. Moving probabilities farther is measurable and survived the control. Improving a prediction, changing a behavior selectively, or producing a reliable repair is a further claim.

That is the lesson I would carry to another model family. Check an attribution against the intervention it estimates. Check a direction against a control that moves the hidden state by the same amount. Then test the intended behavior directly. A faithful measurement of one effect does not automatically resolve the next question.

Appendix: scope, protocol status, and reproducibility

Models and intervention layers

The studies reuse 30 development checkpoints from two node-classification datasets. The narrow GCN has three layers and width 32. The residual GCN has five blocks and width 64. Layer indices below are zero-based hidden-activation indices.

Cell	Checkpoints	Single-layer direction intervention	Primary cross-layer pair
Amazon Computers / narrow GCN	5	0	0 → 1
Amazon Computers / residual GCN	5	0	0 → 1
Roman Empire / narrow GCN	10	0	0 → 1
Roman Empire / residual GCN	10	4	3 → 4

All adjacent pairs were measured in the residual models, but the primary pair anchors the comparisons reported here. Roman Empire's residual single-layer intervention is at the last block, so that layer becomes the target of its primary cross-layer pair. These are different measurement locations, not interchangeable descriptions of one intervention. Analysis uses the specified training and validation scopes, with test labels kept out of these studies.

What the summaries and frozen rules mean

The same checkpoints recur across sections. There are more Roman Empire seeds than Amazon seeds, so a pooled checkpoint median does not weight the four cells equally. Cross-seed comparisons also share model partners. Counts, cell summaries, and scatter plots should not be read as independent replications or confidence intervals.

Active-direction analyses exclude dead components. Count-matched analyses also equalize how many directions are compared across conditions. That distinction arose because some narrow-model channels were inactive while rotated directions were not. For the cross-layer studies, "active channels" means removing dead channels from the edge matrices.

The extensions are exploratory because they reuse inspected development data. Freezing a study's rules before its full evaluation constrains interpretation, but does not turn that evaluation into a new-data confirmation. The individual records preserve the initial probes, amendments, and decision thresholds. The amplification preview and repaired tail rerun are disclosed next to their results because they bear directly on those results' evidential status.

Pooled contrasts are medians of paired checkpoint contrasts. They need not equal differences of pooled medians. Amplification is summarized by taking medians of log2 learned/null ratios within checkpoints and then across checkpoints, before exponentiating. The high-impact subset uses ten features per dictionary, alongside an all-live subset, with three dictionary seeds.

What can be reproduced from the public package

The pinned package contains the decision records, their corrections and retractions, source identities, checkpoint-level figure data, and the figure generator. The companion README gives the regeneration commands. `python3 verify.py` checks the distributed file hashes.

The five figures can be regenerated from the included measurements without loading models or downloading datasets. The extraction checked the hashes of all 30 saved edge matrices and recomputed their signed-effect correlations. The complete extension execution code, checkpoints, and raw matrices are not included in this package. Readers can reaggregate the supplied values and regenerate these figures, but cannot independently replay all extension experiments from this release alone. Hashes identify the archived inputs without substituting for access to them.

AI assistance. I used AI assistance for editorial drafting, source checking, and figure preparation. The distinction between saved measurements and new experiments is explicit throughout. Interpretation of the results remains my responsibility.